

Topic 4: Plant Structure, Function, Biodiversity and Conservation

4.1 – Cell Wall Structure and Key Components

Cell Wall Composition:

- Made up primarily of **cellulose** — a **β -glucose polysaccharide**.
- Contains:
 - **Cellulose microfibrils**: long chains of β -glucose joined by 1,4 glycosidic bonds; hydrogen bonding between chains provides **tensile strength**.
 - **Hemicellulose**: short-chain carbohydrates (e.g. xylose, mannose) that link microfibrils together to reduce flexibility and increase tensile strength
 - **Pectin (calcium pectate)**: binds adjacent cell walls together, especially in the **middle lamella**. Reduces flexibility of overall stem, especially at junctions between cells
 - **Lignin** (in secondary cell walls): adds rigidity and **waterproofs** xylem and makes the vessel hollow by killing the cell, so that there is a continuous column of water flowing without cell components interrupting it

Function:

- Provides **mechanical support** and **maintains cell shape**.
- Resists **osmotic lysis** due to high internal pressure (turgor).
- Allows **intercellular communication** through **plasmodesmata**.

4.2 – EM Identification and Microscopy of Plant Tissues

Using the microscope to identify structures:

- **Transverse sections** of roots, stems, and leaves show:
 - **Xylem**: thick-walled, lignified, often larger vessels.
 - **Phloem**: smaller, sieve tubes + companion cells.
 - **Sclerenchyma**: very thick-walled, supportive tissue, often stained dark.

Electron Microscopy (EM):

- EM allows high resolution to distinguish:
 - **Middle lamella, plasmodesmata, and cell wall thickness.**
 - Distinct **cellulose microfibril patterns** in different layers.

4.3 – Cellulose, Starch and Lignin Structure

Cellulose:

- Polymer of **β -glucose**, straight unbranched chains.
- Alternate glucose units are **inverted (180°)** to allow straight chains.
- Chains held by **hydrogen bonds**, forming **microfibrils**.
- Microfibrils form **fibres**, which give **high tensile strength**.

Starch:

- Polymer of **α -glucose**.
- Two forms: **amylose** (helical, 1,4 bonds) and **amylopectin** (branched, 1,4 and 1,6 bonds).
- **Compact and insoluble** — ideal for energy storage.

Lignin:

- A **complex, non-carbohydrate polymer** found in xylem cell walls.
- Adds **rigidity** and **impermeability to water**.
- Makes the **xylem hollow and strong**, resisting collapse under pressure.

4.4 – Role of Lignin, Xylem and Sclerenchyma in Support

Xylem Vessels:

- Made of **dead, hollow cells** with lignified walls.
- Pits to distribute water pressure equally throughout the stem, avoiding collapse

- Pits also allow water to flow into tissues directly from xylem
- Lignin forms:
 - **Spirals** or **rings** for flexibility and strength.
 - Resists **inward collapse** due to tension from transpiration.
 - Allows **capillary action** and **uninterrupted water flow**.

Sclerenchyma Fibres:

- Also **dead**, elongated cells with **thick, lignified walls**.
- Provide **mechanical strength**, especially in **non-growing regions**.
- Not involved in water transport — purely supportive.

Lignin:

- **Hydrophobic** → prevents water leakage.
- **Mechanical strength** → supports plant against gravity and wind.

Summary Table: Cellulose vs Starch vs Lignin

Property	Cellulose	Starch	Lignin
Monomer	β -glucose	α -glucose	Not a carbohydrate
Bond	$\beta(1\rightarrow4)$	$\alpha(1\rightarrow4)$ and $\alpha(1\rightarrow6)$	N/A
Structure	Straight chains	Helical (amylose), branched (amylopectin)	Irregular, rigid polymer
Function	Cell wall support	Energy storage	Xylem wall strengthening
Solubility	Insoluble	Insoluble	Insoluble
Additional	Forms microfibrils	Compact and branched	Impermeable to water

4.5 – Identification of Xylem, Phloem, and Sclerenchyma Using Light Microscopy

Structures to Identify in Transverse Sections:

Tissue	Key Features Under Light Microscope	Function
Xylem	Large, thick-walled, lignified vessels (often stained red or pink with safranin)	Transports water and minerals; provides support

Phloem	Smaller, thin-walled cells, includes sieve tubes and companion cells Companion cells contain nucleus, ribosomes etc for protein synthesis, mitochondria for ATP formation that is needed for active transport of sucrose, amino acids etc into the cells from phloem	Transports organic solutes (sugars)
Sclerenchyma fibres	Thick, lignified walls; often narrow and elongated	Provides additional strength and support
Parenchyma	Thin-walled, unspecialised cells	Storage, photosynthesis, and packing tissue

Additional Notes:

- **Xylem vessels** are hollow, without end walls, enabling **continuous water flow**.
- **Phloem** contains **sieve plates** between sieve tube elements.
- **Sclerenchyma fibres** are typically located **outside the phloem**.

4.6 – Transport Roles of Sclerenchyma, Phloem, and Xylem

Tissue	Structure	Transport Role
Xylem	Dead cells, lignified walls, no cytoplasm	Passive transport of water and mineral ions via transpiration stream
Phloem	Living sieve tube elements with companion cells	Active transport of organic solutes (e.g., sucrose) through translocation
Sclerenchyma	Dead cells, heavily lignified walls, no role in transport	No transport role — structural only

Mechanisms:

- **Xylem:** Water moves due to **transpiration pull**, **capillarity**, and **root pressure**.
- **Phloem:** Uses **active loading** of sugars into sieve tubes, generating **pressure differences** (mass flow hypothesis).

4.7 – Water and Inorganic Ions in Plant Support

Importance of Water:

- **Turgor pressure** inside plant cells keeps tissues rigid.
- Water enters cells by **osmosis**, generating pressure against cell wall.
- In **non-lignified tissues**, support depends on water.

Inorganic Ions:

Ion	Role
Nitrate (NO_3^-)	For amino acids, proteins, DNA, and growth
Calcium (Ca^{2+})	Strengthens cell walls (as calcium pectate in middle lamella)
Magnesium (Mg^{2+})	Central ion in chlorophyll; essential for photosynthesis
Potassium (K^+)	Regulates stomatal opening; enzyme activation
Phosphate (PO_4^{3-})	Component of DNA, ATP, phospholipids

4.8 – Microscopic Identification of Plant Tissues

4.9 – Uses of Plant Fibres and Comparison with Synthetic Materials

Plant Fibres:

Fibres are **long, strong, cellulose-rich cells** extracted from vascular tissue (e.g. sclerenchyma, xylem) in plants like flax, hemp, cotton, and jute.

Property	Plant Fibres	Synthetic Fibres (e.g. plastics)
Source	Renewable (grown from plants)	Non-renewable (petrochemicals)
Biodegradability	Biodegradable	Non-biodegradable
Strength	High tensile strength (due to cellulose/lignin)	Can vary, often flexible and durable
Sustainability	Can be grown with low input (organic)	Derived from finite fossil fuels
Uses	Rope, fabric, bioplastics, composites	Packaging, textiles, containers

Sustainability and its advantages

- Biodegradable materials
- Carbon neutral production. For example, materials derived from plants release the same amount of carbon dioxide upon burning, as that they had used up during photosynthesis. So there is no net carbon emission

- Usage of renewable resources ensures that the product will be available to future generations
- Reduced contribution to global warming (due to carbon neutrality)

4.10 – Conditions Required for Bacterial Growth

Bacteria reproduce asexually through **binary fission**. Their growth depends on several environmental factors:

Condition	Why It's Important
Temperature	Enzymes work best at optimal temperature (often 25–37°C); extremes denature proteins
pH	Enzyme activity affected by acidity/alkalinity; most prefer neutral pH
Nutrients	Carbon source (e.g. glucose or sucrose), nitrogen (e.g. amino acids, nitrates), minerals
Water availability	Required for metabolic reactions and transport, solvent for enzyme activation and dissolving enzymes
Oxygen availability	Aerobic respiration (obligate aerobes) vs. anaerobic growth (facultative/obligate anaerobes)

In lab settings, these factors are controlled using **agar plates** and **liquid broth cultures**.

4.11 – Investigating Antimicrobial Properties of Plants (Core Practical 7)

Aim:

To investigate how plant extracts (e.g. garlic, mint, tea tree) affect **bacterial growth**.

Method Summary:

1. **Prepare agar plates** inoculated with a **bacterial lawn** (e.g. *E. coli*).
2. Apply plant extract solutions onto sterile paper discs.
3. Place discs onto agar plates **evenly spaced**.
4. Incubate plates (usually at **25–30°C for 24–48 hours**).
5. Measure **zones of inhibition** (diameter in mm).

Variables:

Independent	Type/concentration of plant extract
Dependent	Diameter of zone of inhibition
Controlled	Bacterial strain, incubation time, temperature, volume of extract, disc size

Interpreting Results:

- **Larger inhibition zone = more effective antimicrobial.**
- Compare to **positive control** (e.g. standard antibiotic disc) and **negative control** (e.g. water, ethanol solvent).

Safety and Aseptic Techniques:

- Sterilise all equipment (e.g. via autoclave or alcohol).
- Work near a **Bunsen burner** to reduce airborne contamination.
- Dispose of all bacterial waste in **biohazard bags/autoclave**.
- Burn inoculating loop before use
- Keep lid slightly open to prevent anaerobic conditions which may encourage growth of pathogenic bacteria
- Wash hands with soap and water before and after
- Use non pathogenic bacteria to test so that chances of infections spreading is reduced
- Avoid temperatures above 30 degrees as this may encourage growth of pathogenic strains
- Seal lid using tape to avoid touching petri dish directly and ensure it is sealed properly, except for the tiny part where it remains opened in a small enough gap to allow air with oxygen to enter

◆ 4.12 – Role of Plants in Medicine

Historical and Modern Importance:

- Plants produce **secondary metabolites** to defend against pathogens and herbivores — many of these have **medicinal value**.

Plant Compound	Source	Use
Aspirin	Willow bark	Pain relief, anti-inflammatory
Digitalis	Foxglove	Treats heart conditions
Quinine	Cinchona tree	Antimalarial
Taxol	Pacific yew	Cancer chemotherapy
Morphine	Poppy	Strong analgesic (painkiller)

Key Concepts:

- Many modern drugs are **plant-derived** or inspired by **plant biochemistry**.
- **Sustainable harvesting, biodiversity conservation, and bioprospecting ethics** are crucial to continued drug discovery.

4.13 – Drug Testing: From Historical to Modern Protocols

William Withering and Digitalis Soup:

- In the 18th century, William Withering treated heart failure (dropsy) using **foxglove extract**, which contains **digitalis**.
- His approach was **empirical** and relied on **trial and error**.
- He documented dosages, side effects, and patient outcomes.
- Too little digitalis was ineffective, too much could be fatal — illustrating the importance of **dosage control**.

Modern Drug Testing Process:

Stage	Purpose
Pre-clinical testing	Performed on cells, tissues, and animals to assess toxicity and pharmacological effects. May have ethical implications as animals cannot consent and will undergo side effects, therefore feeling pain
Phase I trials	Involve small numbers of healthy volunteers ; tests safety, side effects, and dosage . Data is analysed by independent scientists Statistical tests may be done
Phase II trials	Conducted on a few hundred patients with the disease to assess efficiency, identify appropriate and effective dose and safety. This must be double blind and data is analysed by statistical tests, showing whether the tested drug is more effective than the current treatment and whether this difference is significant
Phase III trials	Large-scale, randomised, double-blind, placebo-controlled studies on thousands of patients; compares the new drug to existing treatments. Confirmation of effectiveness, confirmed dosage selection and further checking for side effects among large group Use statistical tests (same purpose as phase ii)
Regulatory approval	If successful, the drug is approved by health authorities (e.g., FDA, EMA).

4.14 – Classification and the Three-Domain System

(i) Purpose of Classification:

- Organises the diversity of life into **hierarchical groups** based on **similarities and differences** in phenotype and genotype.
- Aims to **reflect evolutionary relationships**.
- Based on the **species concept**: organisms that can interbreed to produce fertile offspring.

(ii) Evaluating New Data in Classification:

- Advances in molecular phylogeny have allowed use of:
 - **DNA and RNA sequencing**
 - **Amino acid comparisons**
 - **Ribosomal RNA analysis**
- This led to new taxonomic systems, including the:
 - **Three-domain system**: Bacteria, Archaea, and Eukarya
 - Bacteria and Archaea are both prokaryotic but have key biochemical differences.
 - Archaea are more closely related to Eukarya than Bacteria.

Characteristic	Bacteria	Archaea	Eukaryota
Membrane-enclosed nucleus	Absent	Absent	Present
Membrane-enclosed organelles	Absent	Absent	Present
Peptidoglycan in cell wall	Present	Absent	Absent
Membrane lipids	Ester-linked, unbranched	Ether-linked, branched	Ester-linked, unbranched
Ribosomes	70S	70S	80S
Initiator tRNA	Formylmethionine	Methionine	Methionine
Operons	Yes	Yes	No
Plasmids	Yes	Yes	Rare
RNA polymerases	1	1	3
Ribosomes sensitive to chloramphenicol and streptomycin	Yes	No	No
Ribosomes sensitive to diphtheria toxin	No	Yes	Yes
Some are methanogens	No	Yes	No
Some fix nitrogen	Yes	Yes	No
Some conduct photosynthesis	Yes	No	Yes

4.15 – Variety of Life and the Threat of Human Activity

- Over time, life has diversified through **evolution and speciation**.
- Human activities now pose a major threat to biodiversity:
 - **Habitat destruction** (deforestation, urbanisation)
 - **Pollution** (air, water, soil)
 - **Climate change** (temperature shifts, sea level rise)
 - **Overexploitation** (overfishing, hunting, logging)
 - **Invasive species** introduced by humans outcompete natives

These impacts reduce species diversity, disrupt ecosystems, and accelerate extinction rates.

4.16 – Definitions: Biodiversity and Endemism

Biodiversity:

- The variety of life in all its forms:
 - **Species diversity**
 - **Genetic diversity**
 - **Ecosystem diversity**

Endemism:

- A species is **endemic** if it is found **only in one specific geographic location**.
 - Example: Lemurs are endemic to Madagascar.

High levels of endemism usually occur in **isolated ecosystems** (e.g. islands, mountains) and indicate **unique biodiversity**, but also greater **vulnerability to extinction**.

4.17 – Measuring Biodiversity Within a Species

Genetic Diversity via Heterozygosity Index:

Genetic diversity within a population can be quantified using the **heterozygosity index**:

$$\text{Heterozygosity index} = \frac{\text{Number of heterozygotes}}{\text{Total number of individuals in the population}}$$

- A **higher value** indicates **greater genetic variation**, which increases the population's ability to adapt to environmental change.
- Important in conservation: low heterozygosity suggests **inbreeding** and higher extinction risk.

4.18 – Comparing Biodiversity Between Habitats

Simpson's Index of Diversity (D):

$$D = \frac{\sum n(n-1)}{N(N-1)}$$

Where:

- **n** = number of individuals of each species
- **N** = total number of organisms of all species
- **Higher D values** indicate **greater biodiversity**.
- Accounts for both **species richness** (number of species) and **evenness** (abundance of each).

Commonly used in fieldwork to compare different ecosystems.

4.19 – Niche and Adaptation

Niche:

- The **role of a species** within an ecosystem, including:
 - Its **habitat**
 - Its **interactions** (e.g., predation, competition)
 - Its **diet**, behaviour, and abiotic tolerances

Each species occupies a **unique niche**. If two species share the same niche, **competition** results — one may be excluded.

Adaptations:

Type	Definition	Example
Anatomical	Structural features of the body	Thick fur in polar bears
Behavioural	Actions or patterns of activity	Migration in birds
Physiological	Internal processes	Antifreeze proteins in Arctic fish

Adaptations improve **survival and reproductive success**, and arise via **natural selection**.

4.20 – Hardy-Weinberg and Allele Frequency

(i) Using the Hardy-Weinberg Equation:

Used to determine if **allele frequencies** in a population are stable (i.e., not evolving):

$$p + q = 1 \quad (\text{allele frequency}) \qquad p^2 + 2pq + q^2 = 1 \quad (\text{genotype frequency})$$

- **p** = frequency of the dominant allele
- **q** = frequency of the recessive allele
- **2pq** = frequency of the heterozygous genotype

Assumptions of the model:

- Large population
- Random mating
- No mutation
- No migration
- No natural selection

Deviation from expected values indicates **evolutionary change**.

(ii) Causes of Allele Frequency Change:

- **Mutation**: introduces new alleles
- **Natural selection**: favours certain alleles
- **Genetic drift**: random changes, especially in small populations
- **Migration (gene flow)**: introduces or removes alleles

(iii) Reproductive Isolation and Speciation:

- Isolation prevents gene flow between populations:
 - **Geographical** (allopatric)
 - **Behavioural** or **temporal** (sympatric)
- Over time, **mutations accumulate**, and **natural selection** favours different traits → populations become genetically distinct → **new species form**

Definition of Speciation

- **Speciation** is the **evolutionary process** by which **new species arise**.
- A **species** is defined as a group of organisms that can **interbreed to produce fertile offspring**.
- Speciation occurs when **two populations become reproductively isolated** and **accumulate genetic differences** over time due to **mutation, natural selection, and genetic drift**.

Types of Speciation

Type	Description	Cause
Allopatric Speciation	Populations become geographically separated	Rivers, mountains, deserts, habitat fragmentation
Sympatric Speciation	Speciation occurs within the same geographical area	Reproductive barriers (e.g. behaviour, timing, chromosome number)

Examples

- **Allopatric:** A population of insects is split by a mountain range → each group adapts to different conditions and becomes a new species.
- **Sympatric:** Two groups of birds live in the same forest but develop different mating calls → eventually stop interbreeding so they evolve separately
- Different species cannot produce fertile offsprings because gene flow cannot occur from one generation to another. This is because the species have different numbers of chromosomes and therefore cannot combine to undergo meiosis and form gametes

Mechanisms Leading to Speciation

1. Isolation:

- Populations are **prevented from interbreeding**.
- Can be **geographical** (allopatric) or **non-geographical** (sympatric).
- So genes from one group cannot flow to the other

2. Mutation:

- Random mutations introduce **new alleles** in each population.

3. Natural Selection:

- Different **selection pressures** in each population favour different traits.

- Advantageous alleles increase in frequency over generations.
4. **Genetic Drift** (especially in small populations):
- Allele frequencies change **randomly**.
 - Can lead to **magnification (fixation) or loss** of alleles.
5. **Accumulation of Differences:**
- Over many generations, genetic divergence increases.
 - Populations can no longer interbreed even if reintroduced → **new species** have formed.

Types of Reproductive Isolation (Prezygotic and Postzygotic)

Type	Explanation
Temporal isolation	Different mating seasons or times of day
Behavioural isolation	Different courtship behaviours prevent recognition
Mechanical isolation	Physical incompatibility of reproductive organs
Gametic isolation	Gametes cannot fuse due to incompatibility
Postzygotic isolation	Zygote forms but offspring is inviable or sterile (e.g., mule)

(you don't need to know in details)

Evidence of Speciation

- **Ring species:** populations that can interbreed with neighbours but not with distant populations.
- **Fossil records:** show gradual changes in form.
- **DNA analysis:** shows genetic divergence.
- **Hybrids:** reduced fertility in hybrid offspring suggests reproductive isolation.

4.21 – Conservation Methods for Endangered Species

Conservation aims to preserve:

- **Species** and their habitats
- **Genetic diversity** within species
- **Ecosystem function**

In situ Conservation:

- Protects species in their **natural habitat**.

- National parks, wildlife reserves
- Control of invasive species, habitat restoration
- Fencing the natural habitat and imposing laws to prevent human intervention that would cause further destruction
- Spreading awareness of the fact that the species is endangered and the habitat is to be conserved

Ex situ Conservation:

- Protects species **outside their natural habitat**.
 - **Zoos** (captive breeding programmes)
 - **Botanic gardens**
 - **Seed banks** (e.g., Millennium Seed Bank)

Scientific and Practical Approaches:

- **Genetic screening** to maintain genetic diversity
- **Reintroduction programmes** for restoring populations
- **Monitoring and research** on population trends and threats
- **Education and public awareness** to support conservation efforts

Plant Conservation Strategies

1. Botanical Gardens and Arboretums

- **Definition:** Controlled environments where a wide variety of plant species are cultivated for **conservation, research, and public awareness**.

Functions:

- **Conservation of rare and endangered plants**
- Study of **plant physiology, genetics, and adaptive traits**
- **Propagation programmes** to maintain diverse gene pools
- Facilitate **reintroduction** into natural habitats
- Act as **living repositories** for species threatened by habitat destruction or climate change

Arboretums:

- Specialise in **tree species conservation**
- Preserve **genetic diversity** in response to threats like deforestation

Collaboration Between Gardens:

- Exchange of plant materials and data enhances **genetic variation**
- Promotes **educational and awareness programmes** to prevent extinction
- Exchanging seeds, gametes in the form of pollen, plants etc to increase genetic diversity

2. Seed Banks

- **Purpose:** Store seeds under **controlled conditions** to preserve viability long-term.

Process:

1. Seeds are **washed with antimicrobial solution** to prevent microbial decay.
2. **Dried** and examined using **X-rays** to check viability.
3. Stored in **air-tight, labelled containers** at **−18°C to −20°C** to inhibit enzyme activity and germination.

Advantages:

- Stores **large numbers** of seeds from many species in a **compact space**
- **Cost-effective**
- Allows for **replanting and regeneration** when needed

Limitations:

- Some seeds **cannot be stored traditionally**
- Some may **lose viability** or accumulate **mutations**
- **Not all species can germinate** post-storage

3. Cryopreservation

- Used for **seeds or tissues** that cannot be stored using conventional seed bank methods.
- Material is stored in **liquid nitrogen** at **−196°C**.

Benefits:

- Prevents **enzyme activity** and **metabolism**, preserving viability indefinitely
- Maintains **genetic stability**

Drawbacks:

- **Expensive**
- Cannot be used for **all types of plant material**

4. Tissue Culture / Micropropagation

- Cultivation of **plant cells or tissues** in sterile, controlled lab conditions.
- Produces **genetically identical clones** (clonal propagation).

Applications:

- Useful for **rare or slow-growing species**
- Mass production of healthy plants for:
 - **Reintroduction to the wild**
 - **Habitat restoration**
 - **Selective breeding programmes**

However, cloning reduces genetic diversity as all plants will be genetically identical, making the population more susceptible to environmental changes

5. Stud Books and Genetic Management

- **Stud books** are detailed records of:
 - **Lineage**
 - **Genetic traits**
 - **Breeding history**

Benefits:

- Prevents **genetic bottlenecks** and **inbreeding**
- Ensures **high genetic variation**
- Supports **selective breeding** for desirable traits (e.g., disease resistance, drought tolerance)
- Data can be shared between institutions for:
 - **Seed/plant exchange**
 - **Cross-breeding programmes**
 - **Restoration of endangered populations**

6. Secured Zones and Protected Areas

- **Designated areas** such as:
 - **National parks**
 - **Biosphere reserves**
 - **Ecological sanctuaries**

Purpose:

- Protect native plants from:
 - **Deforestation**
 - **Pollution**
 - **Invasive species**
 - **Overgrazing**
 - **Human interference**

Protection Methods:

- **Fences** around vulnerable trees or zones
- Use of **repellents and chemicals** to deter animals/insects
- **Minimal human activity** allowed — strict conservation regulations

Use Case:

- Especially valuable for:
 - **Ancient trees** (e.g., 400+ years old)
 - **Fragile or ecosystem-sensitive species**
 - Species that **cannot be cultivated** in artificial gardens

7. Artificial Pollination

Artificial pollination is the **manual transfer of pollen** from the anther of one flower to the stigma of another using **human intervention** (e.g. by hand, brush, or instruments). It is an intentional form of **cross-pollination**.

Advantages and Purpose:

Benefit	Explanation
Controlled cross-pollination	Humans select specific parent plants using stud books to ensure desirable traits are passed on.
Maximises genetic variation	Carefully chosen pollen sources increase allele diversity in offspring. Preventing self pollination
Improved fertilisation success	Unlike natural pollination (which depends on chance and pollinator activity), artificial pollination ensures high rates of both pollination and fertilisation .
No reliance on pollinators	Useful when natural pollinators are absent or unavailable. Prevents reliance on insects or wind. Allows to pollinate in absence of other flowers or plants of same species around, as this would slow down pollen transfer
Supports population growth	Ensures pollination of every flower , particularly in species with short-lived flowers that may wilt before natural pollination occurs.
Enables selective breeding	Pollen can be tested and selected based on desired traits such as disease resistance or adaptability.

When is Artificial Pollination Useful?

- **Rare or endangered plants** with low natural pollination success.
- **Short flowering duration** – when flowers wilt too quickly for insect pollinators to act.
- **Conservation and restoration** programmes where population growth and genetic control are crucial.
- **Commercial agriculture** for increasing yield and uniformity of traits.

Conservation of Endangered Animals – Role of Zoos

1. Zoos as Conservation Centres

Controlled Environment:

- Provide **safe, enclosed habitats** that protect animals from **predators, poaching, and food shortages**.
- Allow for **close monitoring** and **environmental control**, tailored to each species' needs.

Scientific Research:

- Study of **animal behaviour, reproduction, and physiology** under controlled conditions.

- Enables optimisation of husbandry practices and improves **captive breeding success**.

Public Education and Awareness:

- Raise awareness about endangered species and conservation threats.
- Increase **public engagement, support, and funding** for conservation efforts.

2. Genetic Management and Selective Breeding

Stud Books:

- Used to document each individual's:
 - Genetic traits
 - Lineage
 - Breeding history
- Help identify ideal breeding pairs to **avoid inbreeding** and ensure **maximum genetic diversity**.

Selective Breeding:

- Parent animals chosen based on **health, genetic diversity, and adaptive traits**.
- Increases offspring **fitness** and **resistance to selection pressures** (e.g. disease, climate change).

3. Assisted Reproductive Technologies

Cryopreservation:

- Sperm, eggs, and embryos are frozen in **liquid nitrogen (−196°C)** to preserve genetic material long-term.
- Ensures gametes can be shared between zoos or used in the future.

Artificial Insemination (AI):

- Sperm is inserted directly into the female's reproductive tract.
- Useful when:
 - Animals don't mate naturally in captivity
 - Individuals are located in different zoos
 - Selective breeding

In Vitro Fertilisation (IVF):

- Sperm and egg are fertilised in a lab setting.

- Produces **viable embryos** that can be implanted into:
 - The **genetic mother**, or
 - A **surrogate female**
- Offers higher fertilisation success rates and more controlled reproduction.
- Allows even immotile sperm to fertilise eggs as the sperm cells don't have to swim to the egg

4. Gene Banks and Zoo Collaboration

- Zoos contribute to **gene banks**:
 - Store **sperm, eggs, and embryos** to preserve genetic diversity.
- Exchange of:
 - **Animals**
 - **Gametes (e.g. sperm)**
 - Helps combat **genetic drift** and maintain a **diverse gene pool** across institutions.

5. Reintroduction Programmes

Goals:

- Restore animals to **natural or semi-natural habitats**.
- Increase population sizes in the wild.
- Increase biodiversity in natural habitats

Stages:

1. **Reinforce natural behaviours:**
 - Reduce human contact
 - Limit feeding to encourage **hunting, foraging, or grazing**
2. **Select and prepare suitable habitat:**
 - Protect area from **human disturbance** or **predators**
 - Ensure availability of **food, shelter, and breeding sites**
3. **Community involvement:**
 - Educate local populations
 - Promote **conservation-friendly attitudes**

- Involve locals in habitat protection

Strategy	Purpose
Controlled environment	Ensures safety and facilitates research
Stud books and breeding	Increases genetic health of offspring, increases genetic diversity in offsprings and ensures the offspring inherits adaptive traits
AI and IVF	Boost reproductive success, reduce limitations of infertility or lack of mating behaviour
Cryopreservation	Preserves genetic material across time and space
Zoo collaboration	Maximises gene flow and genetic diversity
Reintroduction	Establishes sustainable wild populations

Issues with Zoos in Conservation

While zoos play a major role in the conservation of endangered species, they also face several limitations that can affect the **ethical, genetic, behavioural, and ecological success** of their efforts.

1. Altered or Unnatural Animal Behaviour

- Animals in captivity may **not behave as they would in the wild**.
- Lack of **predators, hunting, and migration** opportunities results in:
 - **Stereotypical behaviours** (e.g. pacing, overgrooming)
 - **Reduced stimulation**, leading to stress and poor welfare
- Behavioural changes can hinder **successful reintroduction** into the wild, making the species more vulnerable and susceptible to being wiped out, reduced in numbers or even extinction

2. Small Populations and Inbreeding

- Many zoo populations are **genetically limited** due to inbreeding within small group
- Breeding within a **small gene pool** can lead to:
 - **Inbreeding depression**
 - Reduced fertility and weakened immune systems
 - **Increased risk of harmful mutations**
- Stud books and genetic management aim to minimise these effects, but **complete genetic diversity is hard to maintain**.

3. Mating Challenges in Captivity

- Some animals **refuse to mate** in captive environments.
- Chosen partners may be **rejected**, reducing breeding success.
- Assisted reproductive techniques (ART) such as:
 - **Artificial insemination (AI)**
 - **In vitro fertilisation (IVF)** are used to **increase fertilisation rates** and bypass natural breeding reluctance.

4. Ethical Concerns

- Keeping wild animals in captivity for breeding or public display raises **moral and ethical questions**:
 - Is it acceptable to confine animals for human benefit?
 - Do enclosures adequately replicate natural habitats?
- Welfare concerns arise when animals are kept in **unnatural settings** or with **inadequate stimulation**.
- Critics argue that **conservation should occur in situ** (in natural habitats), not behind bars.

5. Difficulty with Reintroduction to the Wild

- Many zoo-bred animals **struggle to survive after reintroduction** due to:
 - Lack of **natural instincts** (e.g. hunting, predator avoidance)
 - **Dependence on humans** for food and care
 - Poor ability to **find food, avoid predators, or compete for mates**
 - Greater **susceptibility to disease** and stress in wild environments
- Reintroduction efforts often require **long-term preparation**, including:
 - Behavioural training
 - Habitat protection
 - Community education

6. Entertainment vs Conservation

- Some zoos **prioritise public entertainment and profit** over genuine conservation.

- Focus on "**flagship species**" (e.g. pandas, tigers) instead of **critically endangered or lesser-known species**.
- Poorly designed enclosures and lack of enrichment lead to **animal stress and poor mental health**.
- Ethical zoos are shifting toward **education, enrichment, and welfare-focused design**, but this is not universal.

7. Lack of Resources and Focus

- Not all zoos have the **funding, expertise, or infrastructure** to contribute meaningfully to conservation.
- Many zoos focus on **popular or charismatic species**, ignoring **lesser-known but critically endangered organisms**.
- Smaller or underfunded zoos may:
 - Fail to meet modern welfare standards
 - Lack capacity for **proper genetic management or breeding programmes**